

Unit 13, Lesson 5: Weighted Averages



Benchmark

MA.912.GR.3.1 Determine the weighted average of two or more points on a line.



Learning Targets

- ☐ I can determine the weighted average of two or more points on a line.
- ☐ I can explain how changing the weight moves the weighted average of points.



13.5.1 Warm-Up: Between Two Numbers

A stock exchange, or stock market, is a system for buying and selling bonds, securities, or stocks. A stock is a share in the ownership of a company. NASDAQ is short for National Association of Securities Dealers Automated Quotations, which is the world's largest electronic stock exchange. Each company that is traded has an abbreviation of one to four letters.

Jaxon's grandmother bought 100 shares of Netflix and 100 shares of Apple for him when he was born on March 29, 2003. She paid \$1.37 per share for Netflix (NASDAQ: NFLX) and paid \$0.21 per share for Apple (NASDAQ: AAPL).

On Jaxon's 18th birthday, Netflix's stock price was at \$513.95 and Apple's stock price was at \$121.39.

1. How much were the stocks worth on Jaxon's 18th birthday?

The stock price history of Netflix shows that at one point the stock had increased in worth by 45,000%, and Apple stock had increased in worth by 140,000%.

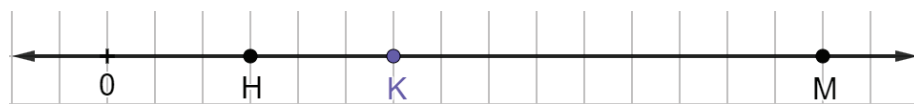
2. Write the decimal equivalent of each percentage increase.

Stock	Decimal Equivalent
Apple	
Netflix	



13.5.2 Exploration: Exploring Weighted Averages

1. A number line is shown. Each grid space is 1 unit.



- a. Find the midpoint of $\overline{HM}$ and label it B .

Assuming that points H and M have the same weight, the diagram represents how points H , B , and M are in balance.



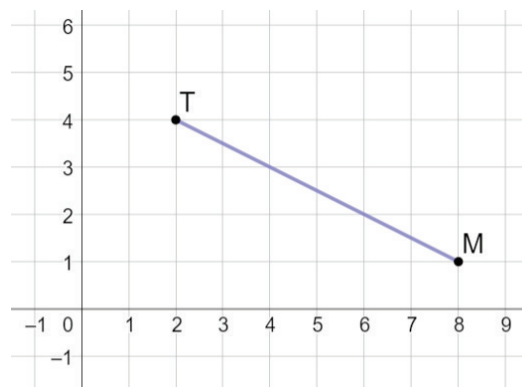
- b. Discuss with your partner why point B acts as a point of balance. Summarize your discussion.
- c. Keyshawn told his partner that if the scale is drawn where the scale is centered on point K , then point H would be higher than point M . Describe or draw what would have to be done to balance the scale when it is centered at point K .

The ratio of $HK:KM$ is 3:9, which can be written as $\frac{1}{3}$. The ratio, $\frac{1}{3}$, can be used to determine the location of point K using the steps shown.

$$\frac{3}{4}(\text{point } H) + \frac{1}{4}(\text{point } M) = \frac{3}{4}(3) + \frac{1}{4}(15) = \frac{9}{4} + \frac{15}{4} = \frac{24}{4} = 6$$

The ratio $\frac{1}{3}$ was used to find $\frac{3}{4}$ and $\frac{1}{4}$ by noting that the ratio indicates there are a total of 4 parts. To balance at point K , point H would be "heavier" than point M , so it is weighted as 3 of the 4 parts with point M weighted as 1 of the 4 parts.

2. Point T has coordinates $(2, 4)$, and point M has coordinates $(8, 1)$.



- a. What is the midpoint of line segment TM ?

b. Complete the table.

	<i>x</i> -value	<i>y</i> -value
<i>T</i> (2, 4)	$\frac{1}{2}$ ()	$\frac{1}{2}$ ()
<i>M</i> (8, 1)	$\frac{1}{2}$ ()	$\frac{1}{2}$ ()
Total		

- c. Discuss with your partner how the table shows the process of finding the midpoint.
- d. Complete the expression that can be used to find the point that partitions the line segment in a 1:1 ratio. *T* represents the coordinates of point *T*, and *M* represents the coordinates of point *M*.

$$\frac{1}{\quad}T + \frac{1}{\quad}M$$

- e. Find the point that partitions directed line segment *TM* in a 2:1 ratio.

f. Complete the table to calculate $\frac{1}{3}T + \frac{2}{3}M$.

	<i>x</i> -value	<i>y</i> -value
<i>T</i> (2, 4)	$\frac{1}{3}$ ()	$\frac{1}{3}$ ()
<i>M</i> (8, 1)	$\frac{2}{3}$ ()	$\frac{2}{3}$ ()
Total		

- g. What do you notice about the total in the table and the point found in part e that partitions the directed line segment in a 2:1 ratio?

3. Line segment *KR* has endpoints at *K*(−6, 1) and *R*(2, 5).

- a. Complete the expression for the point that partitions directed line segment *KR* in a 3:1 ratio.

$$\frac{\quad}{4}K + \frac{\quad}{4}R$$

- b. Use the expression to find the point that partitions directed line segment *KR* in a 3:1 ratio.



13.5.3 Guided Instruction: Determining Weighted Averages

Weight is a measure of force pushing or pulling on an object due to gravity. This means that finding a weighted average puts more “weight” on one point than another.

If B is weighted twice as heavily as A , then during the calculation, B is to measure two times the value of A . In other words, B has two times the “gravitational pull” of A .

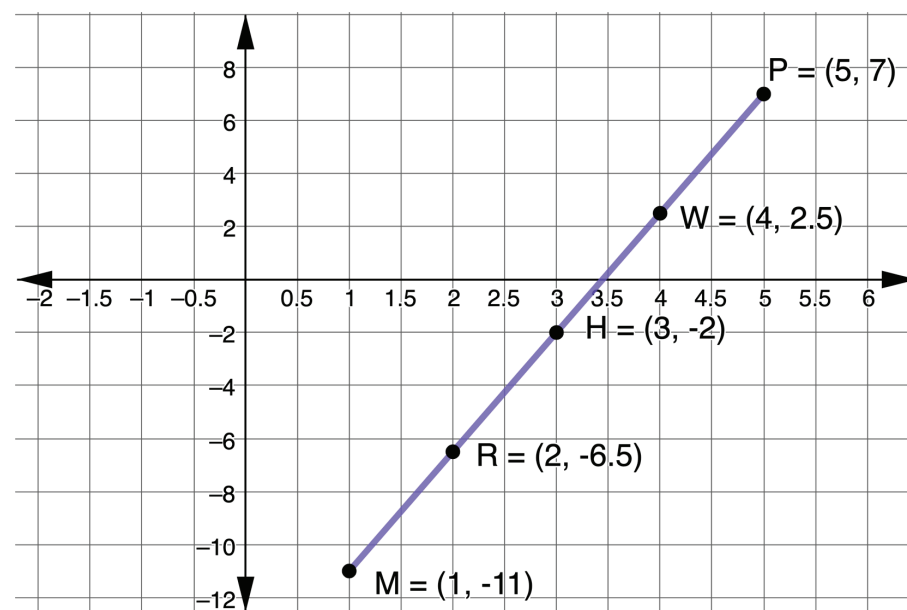
Think of it in terms of a seesaw. In the picture below, point B is weighted more heavily than point A .



1. Discuss with your partner what the girls close to point B would have to do so that the seesaw would balance. Summarize your discussion.



Directed line segment $\overline{MP}$ is shown, where point M is at $(1, -11)$ and point P is at $(5, 7)$. Point H is the midpoint of $\overline{MP}$, point W is the midpoint of $\overline{HP}$, and point R is the midpoint of $\overline{MH}$.



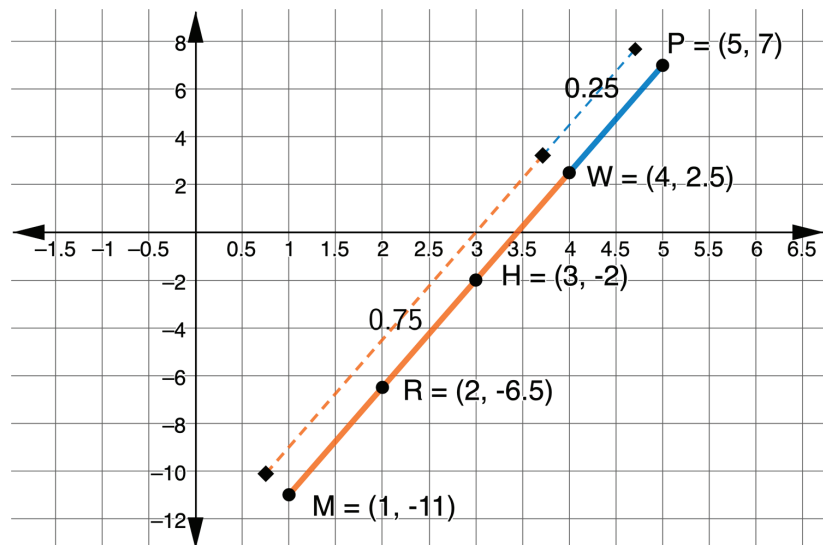
Since point H is the midpoint of $\overline{MP}$, the coordinates of point H can be determined by finding the average of points M and P .

$$\frac{M+P}{2}$$

Point W partitions the line segment $\overline{MP}$ so that point P weighs three times as much as point M . This means that point W is closer to point P because point P is “heavier” than point M . The given expression can be used to calculate this weighted average when point P weighs three times as much as point M .

$$\frac{M+P+P+P}{4}$$

The graph shows the weighted averages.



2. Discuss with your partner why the distance from point W to point P is shorter if point P is "heavier" than point M . Summarize your discussion.

3. Point R partitions the line segment MP so that point M weighs three times as much as point P . Write an expression that shows this weighted average.



A midpoint calculation weighs points on a line segment equally. In a **weighted average** calculation, points on a line segment can be weighted differently.

4. Write the expression that can be used to find the point that partitions $\overline{MP}$ in the given ratio, and find the endpoint that is the weighted point.

Ratio	Expression	Point	Weighted Point
1:1			
1:3			
3:1			

5. A line segment is shown in the table with three different weighted averages. Write the expression that can be used to find each weighted average.

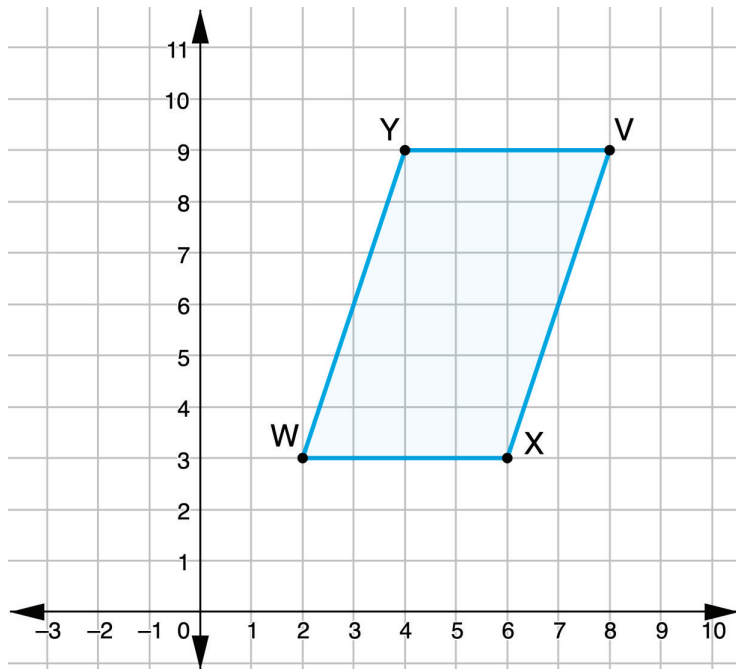
Partition	Graph	Expression
4:1		
2:3		
1:5		

6. Discuss with your partner how changing the weights moves the weighted average on the number line. Summarize your discussion.



13.5.4 Collaboration: Weighted Averages and Dilations

1. $VXWY$ has vertices at $V(8,9)$, $X(6,3)$, $W(2,3)$, and $Y(4,9)$.

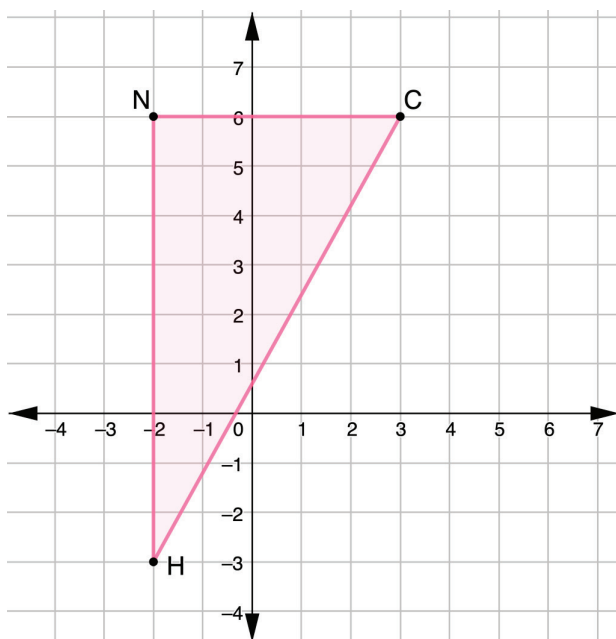


- Find the point that partitions $\overline{WX}$ in a 1:4 ratio. Label it R .
- Find the point that partitions $\overline{WY}$ in a 1:4 ratio. Label it T .

- Find the point that partitions $\overline{VV}$ in a 1:4 ratio. Label it S .

- Is $WTSR$ a dilation of $WYVX$? Justify your answer.

2. Triangle NCH has vertices at $N(-2,6)$, $C(3,6)$, and $H(-2,-3)$.



Use segment partitioning to dilate $\triangle NCH$ using center C and scale factor $\frac{3}{4}$.



13.5.5 Your Turn!

1. Line segment MK has endpoints at $M(-2, 1)$ and $K(8, 7)$.
 - a. Write an expression for the point that partitions directed line segment MK in a 2:3 ratio.
 - b. Use the expression to find the point that partitions directed line segment MK in a 2:3 ratio.



13.5.6 Wrap-Up: Ticket Out the Door

1. A number line is shown. Each grid space is 1 unit.



- a. Determine the point on the number line that satisfies the expression shown. Label the point M .

$$\frac{2}{5}(\text{point } W) + \frac{3}{5}(\text{point } T)$$

- b. What is the ratio $WM:MT$?

Unit 13, Lesson 6: Classifying Triangles



Benchmark

MA.912.GR.3.2 Given a mathematical context, use coordinate geometry to classify or justify definitions, properties, and theorems involving circles, triangles, or quadrilaterals.



Learning Targets

- ☐ I can use coordinate geometry to classify triangles in a mathematical or real-world context.
- ☐ I can draw a triangle on a coordinate plane using given definitions, properties, or theorems.